Study of Radioactive Source Monitoring Based on ElasticNet Regression Method

Bachelor's Thesis Defense Report

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April 2026

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Research Background

Nuclear Facility Decommissioning

- Global decommissioning projects increasing (Fukushima Daiichi, etc.)
- IAEA: Accurate source distribution is prerequisite for optimized decommissioning

Three-Dimensional Complexity

- **Composite sources**: localized hot-spots + continuous activation + dispersed media
- **Nuclide diversity**: ^{134}Cs, ^{137}Cs, ^{60}Co, ^{131}I
- **Constraints**: limited detectors, time, dose limits

Core Challenge

- The inverse reconstruction problem is both **underdetermined** ($m \ll n$) and **ill-posed**
- A single method must handle sparse hotspots, continuous regions, and multi-nuclide discrimination

Problem Statement

Definition 1.1 Mathematical Formulation

$$\mathbf{d} = A \cdot \mathbf{s} + \varepsilon$$

d: measurements A: contribution matrix **s**: source vector ε: noise

Key Challenges

- **Underdetermined**: fewer detectors than unknowns ($m \ll n$)
- **Ill-posed**: noisy data leads to unstable solutions
- **Mixed**: sparse hot spots + continuous activation regions

Existing Methods and Limitations

Method	Sparse	Continuous	Multi-n.	Speed	Core Limitation
ML-EM/MPMI	Fair	Good	Poor	Slow	$O(n^3)$; noise sensitive
IRT-DEGL	Fair	Fair	Fair	V.Slow	Global optimization cost
LASSO	**Exc.**	Poor	Good	Fast	Over-compresses continuous
Fused LASSO	Fair	Good	Good	Med.	Needs neighborhood graph
ElasticNet	**Exc.**	**Exc.**	Good	Fast	Not yet applied to nuclear

Table 1: Comparison of radioactive source reconstruction methods

Why ElasticNet?

Advantages over LASSO and Fused LASSO

- LASSO: over-compresses continuous regions to discrete points
- Fused LASSO: needs predefined neighborhood graph + ADMM convergence issues
- ElasticNet: L1+L2 synergy without extra spatial constraints

Cross-Domain Evidence

- Nuclear medicine (SPECT): outperforms pure L1, L2, fixed-weight EN
- XLCT imaging: superior localization + dual-source resolution
- Fluorescence molecular tomography: 30%+ error reduction

Theorem 2.1 Research Gap

ElasticNet has **not been applied** to nuclear radioactive source monitoring — this thesis fills that gap.

ElasticNet Regression Formulation

Definition 3.1 Objective Function (Zou & Hastie, 2005)

$$\min_{\mathbf{w}} \quad \|\mathbf{y} - X\mathbf{w}\|_2^2 + \alpha \left[\rho \|\mathbf{w}\|_1 + \frac{1-\rho}{2} \|\mathbf{w}\|_2^2 \right]$$

L1 vs. L2 Penalty Roles

- **L1**: promotes sparsity $\rightarrow$ hot-spot localization
- **L2**: groups correlated variables, suppresses oscillation $\rightarrow$ continuous reconstruction
- $\rho \rightarrow 1$: LASSO; $\rho \rightarrow 0$: Ridge

Key Properties

- Strictly convex when $\lambda_2 > 0 \rightarrow$ unique global minimum
- Solved by Coordinate Descent; no ADMM convergence issues

Physical Interpretation

L1: Sparse Hot-Spot Localization

- Count-type penalty on "how many grid cells contain sources"
- Drives irrelevant coefficients to exact zero

L2: Continuous Region Smoothing

- Suppresses individual amplitude, promotes neighboring cells to have similar values
- Preserves continuous region morphology (not "discrete spots" like LASSO)

Theorem 3.1 Synergy Effect

Near hotspots: L1 dominates $\rightarrow$ precise localization. Inside continuous regions: L2 dominates $\rightarrow$ smooth morphology. Transition zones: both cooperate.

Algorithm Pipeline

Stage 1: Preprocessing

- Column normalization: scale each column of A to unit L_2 norm
- Non-negative SVD matrix construction
- Condition number assessment for adaptive α range

Stage 2: ElasticNet Optimization

- `ElasticNet(positive=True)` — hard non-negativity constraint
- GridSearchCV: α (50 values), ρ (80 values), 5-fold CV
- Custom intercept penalty for $|\beta_0/\bar{y}| > 10\%$

Stage 3: Post-processing

- Denormalize: $\mathbf{w}_{\text{est}} = \mathbf{w}_{\text{norm}} \oslash \text{col_norms}$
- Non-negative truncation, intercept analysis, evaluation metrics

Key Optimization: Column Normalization

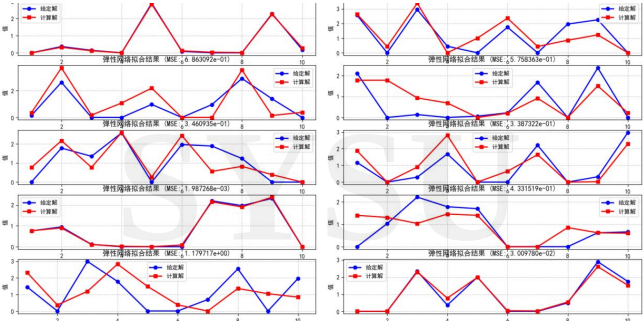

Figure 1: Column normalization: cond=100, 5 runs all achieve MSE=10^{-4} to 10^{-9}

Non-negative SVD + Non-negative ElasticNet

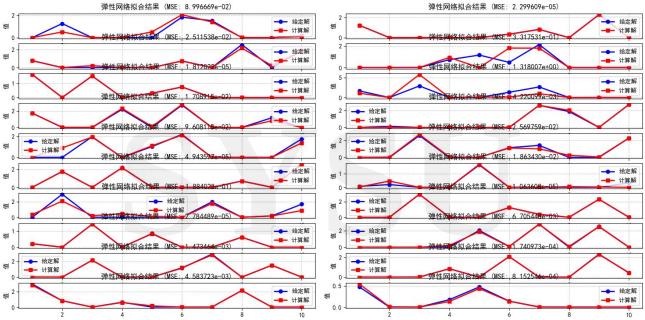

Figure 2: Non-negative SVD + non-negative EN: Exp1/Exp3 reach MSE=10^{-9}

Intercept Fitting Analysis

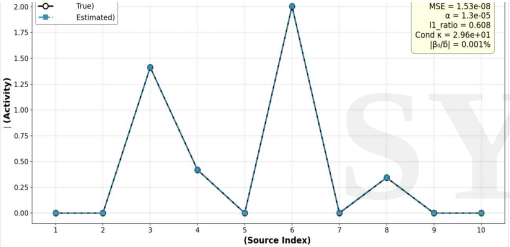

Figure 3: Good case: small intercept

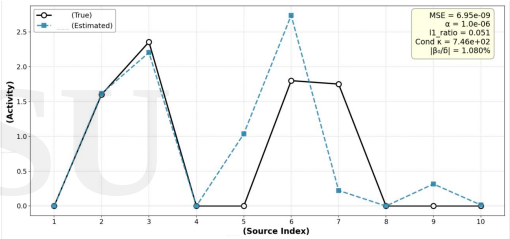

Figure 4: Bad case: large intercept

PHITS Simulation Configuration

PHITS (JAEA): Monte Carlo Particle Transport

- 10^8 particles $\times$ 100 batches; Cs-137, 0.662 MeV γ-ray

Parameter	Value
Simulation space	Spherical (R=10m, reactor-centered)
Radioactive sources	10 Cs-137 point sources, 1 Ci each
Detectors	8 spherical detectors (R=25mm)
Grid division	$4 \times 3 = 12$ cells
Contribution matrix	8×12 (only 11/96 non-zero)
Matrix rank	6/8

Table 2: PHITS simulation parameters

Source-Detector Geometry

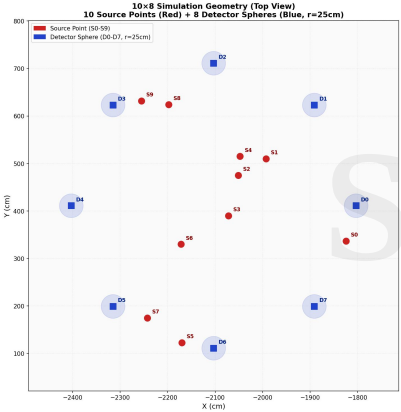

Figure 5: 10 sources → 12 cells

- 10 Cs-137 point sources (1 Ci)
- $4 \times 3 = 12$ grid cells
- 8 spherical detectors (R=25mm)
- 7 active cells, 5 empty
- Matrix: 8×12, rank=6/8

PHITS Validation Results

Best: $\alpha = 10^{-6}$, `l1_ratio` $= 0.99$, $R^2 = 0.463$

7 non-zero cells detected, 5 with $< 2\%$ error

Cell	True (GBq)	Estimate (GBq)	Error
Cell 1	1.265×10^5	1.265×10^5	**0.0%**
Cell 5	5.266×10^4	3.93×10^2	99.3%
Cell 6	1.192×10^5	2.31×10^3	98.1%
Cell 7	5.894×10^4	5.894×10^4	**0.0%**
Cell 8	5.724×10^4	5.724×10^4	**0.0%**
Cell 9	4.870×10^4	4.870×10^4	**0.0%**
Cell 10	1.352×10^5	1.373×10^5	**1.2%**

Reconstruction Visualization

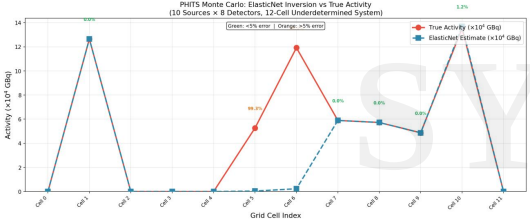

Figure 6: True vs. estimated distribution

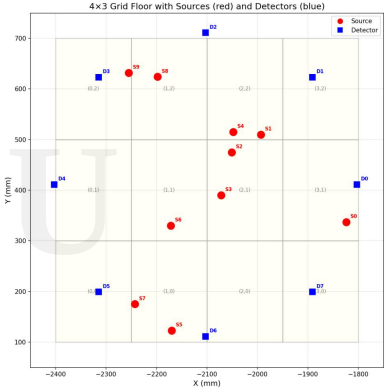

Figure 7: Contribution matrix heatmap

Error Source Analysis

Cell 5 & 6 Failure — Physical, Not Algorithmic

- Cell 6: only Detector 1 responds, weak signal (10^{-9} level)
- Cell 5: only Detector 5 responds, weak signal (10^{-10} level)

Theorem 6.1 Matrix Rank Analysis

Rank of $A = 6/8$. Only 6 of 12 cells algebraically distinguishable. Physical indistinguishability from detector geometry, **not algorithm failure**.

Implication

Improved detector layout would resolve these errors. ElasticNet is NOT the bottleneck.

Condition Number Impact

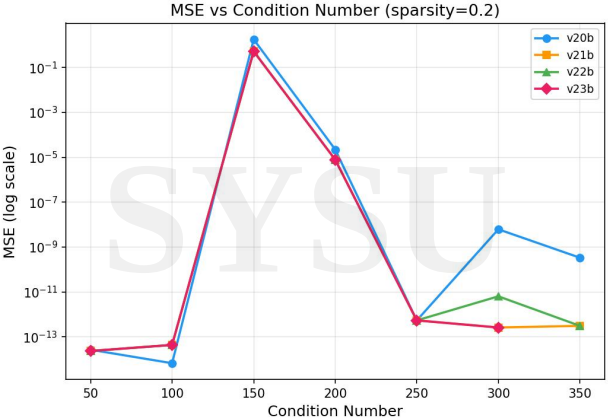

Figure 8: Comparison of adaptive CV, segmented search, and fusion strategies

Bayesian Optimization Results

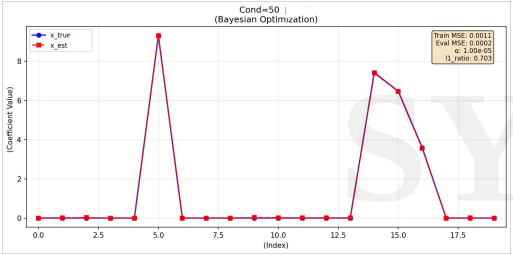

Figure 9: Bayesian opt.: sparse source

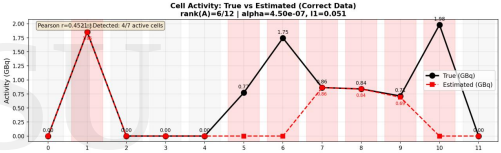

Figure 10: Bayesian + PHITS validation

Grid Search vs. Bayesian Optimization

Difficulty Inflection Points at Cond ≈ 150 and 300

- Not "more ill-conditioned = harder" — critical spectral structure matters
- Optimal `l1_ratio` ≈ 0.99 universally — confirms strong sparsity prior

	Grid Search	**Bayesian Opt.**
Evaluations	150–450	40
Speed	Baseline	5–10× faster
Accuracy (low cond)	Reference	Comparable
Accuracy (high cond)	**Higher ceiling**	Slightly worse

Table 3: Complementary, not substitutive

Conclusions

1. ElasticNet fills the application gap

Effectively handles “sparse + continuous” composite source reconstruction.

2. Optimization strategies improve accuracy

Column normalization + adaptive α + non-negative constraint + intercept penalty.

3. PHITS validation confirms practical value

All 7 non-zero cells detected; 5 cells with $< 2\%$ error.

4. Bayesian optimization: complementary to grid search

70–90% computation savings; grid search has higher accuracy ceiling.

Future Work

Mobile Detection Platforms (UGV / UAV)

- Autonomous patrol → richer contribution matrix → improved stability

Algorithm Improvements

- Adaptive ElasticNet for automatic variable selection
- Bayesian ElasticNet with uncertainty quantification
- Deep learning + ElasticNet model cascade

Extended Applications

- Multi-energy source discrimination & 3D volumetric reconstruction
- Real-time online reconstruction system

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Questions & Discussion

Thank you for your attention!

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